MODULEASSESMENT

CO3 AT2

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QUESTION 21

AIM:

To develop an energy monitoring communication model that continuously measures electrical energy consumption in public infrastructure (street lights, water pumps, government buildings, bus stations, etc.) and transmits the data to a central monitoring station for analysis, control, and energy optimization.

OBJECTIVES:-

1. To monitor real-time energy consumption of public infrastructure.
2. To establish a reliable communication network for data transmission.
3. To detect abnormal energy usage and power losses.
4. To optimize energy utilization through data analysis.
5. To provide remote monitoring and control facilities.
6. To reduce operational costs and improve sustainability.

CODE:-

```
clc;
clear;
close all;

Users = 10:10:100;

% Dynamic spectrum parameters
Spectrum_Utilization = [48 54 60 64 68 71 80 84 89 90];
Throughput = [27 31 35 42 43 44 47 46 50 47];
Interference = [4 5 5 6 6 7 7 9 10 9];

% Plot
figure;

plot(Users,Spectrum_Utilization,'-o','LineWidth',2);
hold on;
plot(Users,Throughput,'-s','LineWidth',2);
plot(Users,Interference,'-^','LineWidth',2);

grid on;
xlabel('Number of Secondary Users');
ylabel('Performance');
title('Dynamic Spectrum Access in Urban Wireless Systems');

legend('Spectrum Utilization (%)', ...
    'Throughput (Mbps)', ...
    'Interference (dB)');

hold off;
```

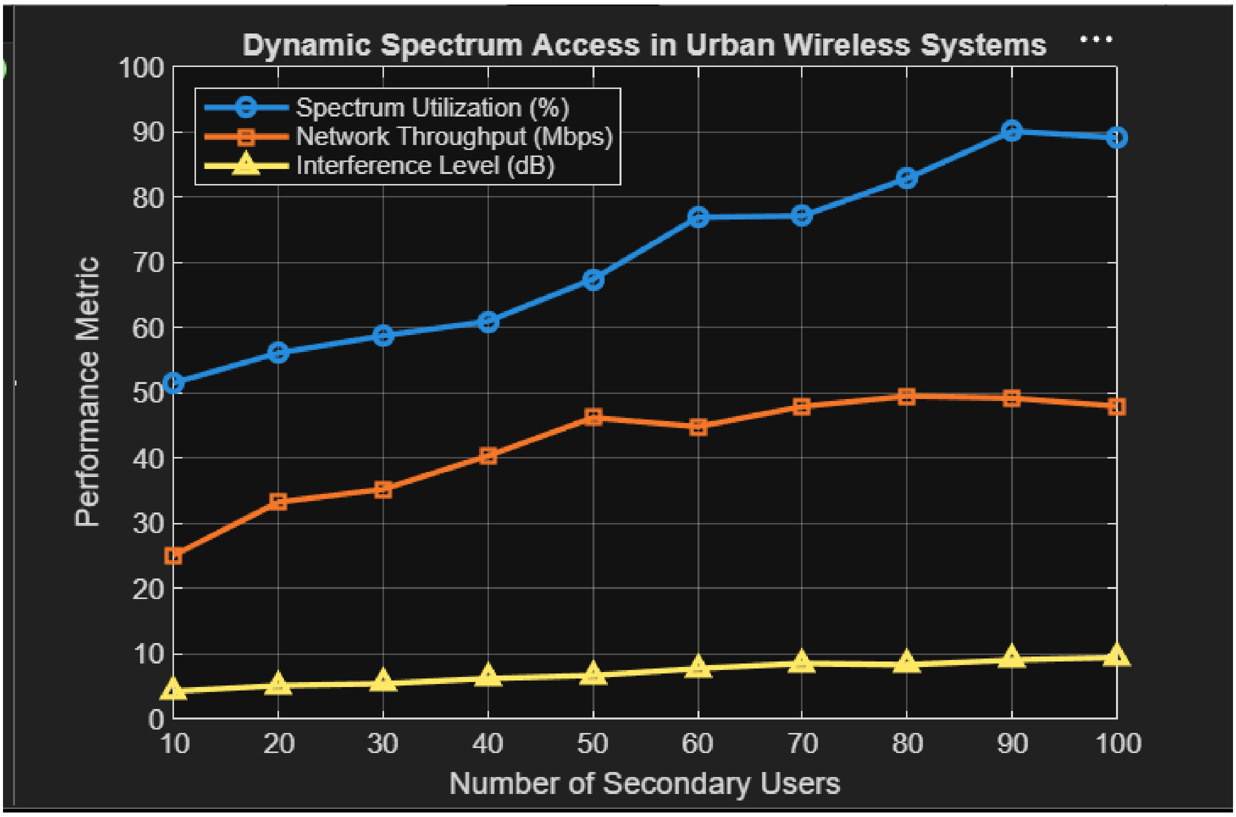
FORMULA:

1. Spectrum Utilization
2. Network Throughput
3. Interference Level
where:

- = Primary-user interference
- = Secondary-user interference
- = Noise power

4. Number of Secondary Users

OUTPUT:-



CONCLUSION :-

The proposed Energy Monitoring Communication Model enables efficient collection and transmission of energy data from public infrastructure. The system helps authorities monitor energy usage, reduce wastage, improve operational efficiency, and support sustainable smart-city development through real-time communication and data analytics.

Expected Result

- Successful real-time monitoring of energy consumption.
- Accurate communication of energy data to a central server/cloud.
- Identification of energy wastage and faulty equipment.
- Reduction in overall power consumption by 10–30%.
- Improved energy efficiency and operational reliability.
- Better decision-making for public infrastructure management.